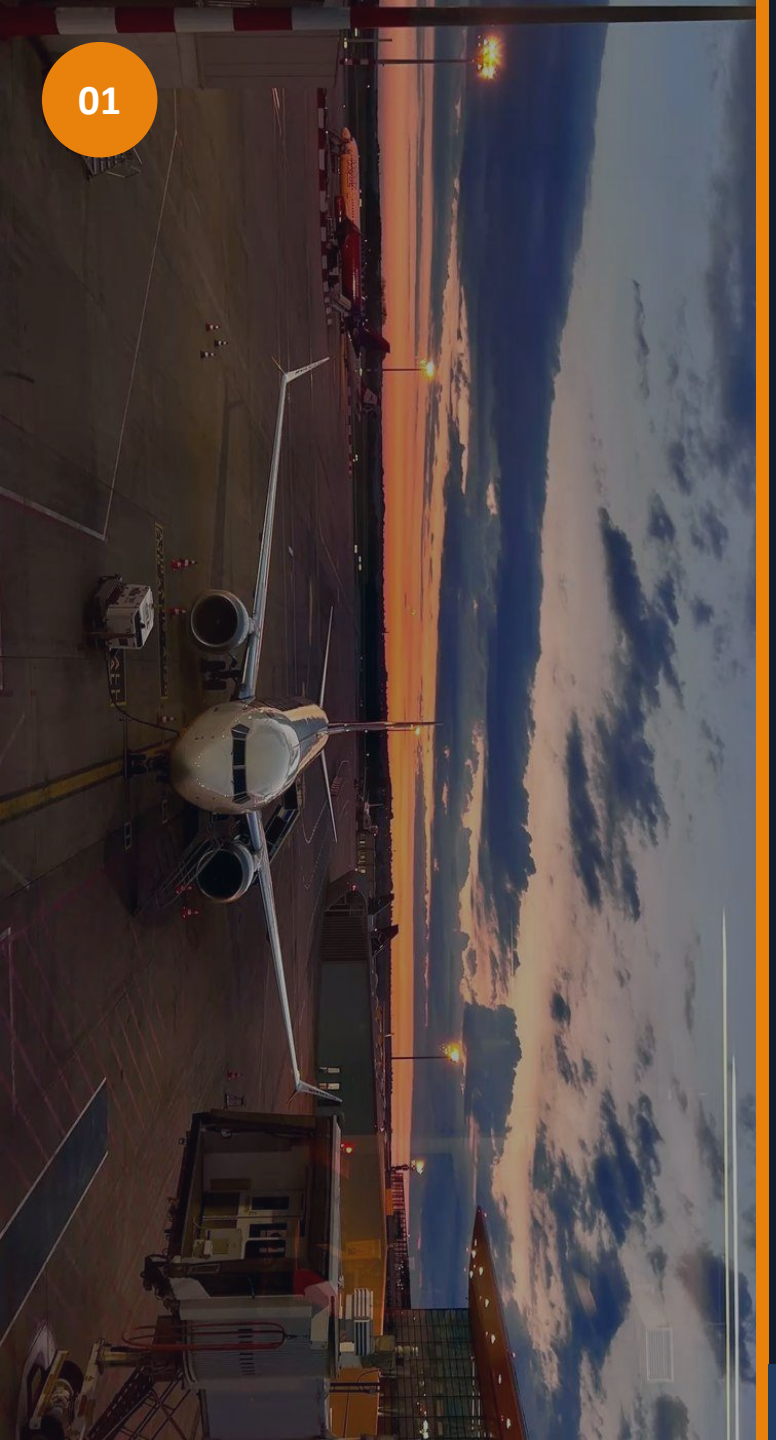
Airline Operations and Revenue Analysis Using SQL

Portfolio SQL Case Study

DATA ANALYSIS | SQL | AVIATION



01

Total Revenue Analysis

₹1,83,08,869

Total Revenue Generated

Aggregated ticket prices and baggage fees across all flight bookings to compute total operational revenue.

SQL QUERY

```
SELECT ROUND(SUM(ticket_price + IFNULL(baggage_fee,0)),2) total_revenue
FROM fact_bookings;
```

RESULT

	total_revenue
▶	18308869

Total Bookings

4,169

Bookings Recorded

Counted all transactions in the fact bookings table to measure overall demand volume across the network.

SQL QUERY

```
SELECT COUNT(*) total_bookings
FROM fact_bookings;
```

RESULT

Result Grid

	total_bookings
▶	4169

03

Highest Revenue Airline

₹3,805,500

Vistara — Top Performer

Grouped bookings by airline and summed revenue to identify Vistara as the leading revenue generator.

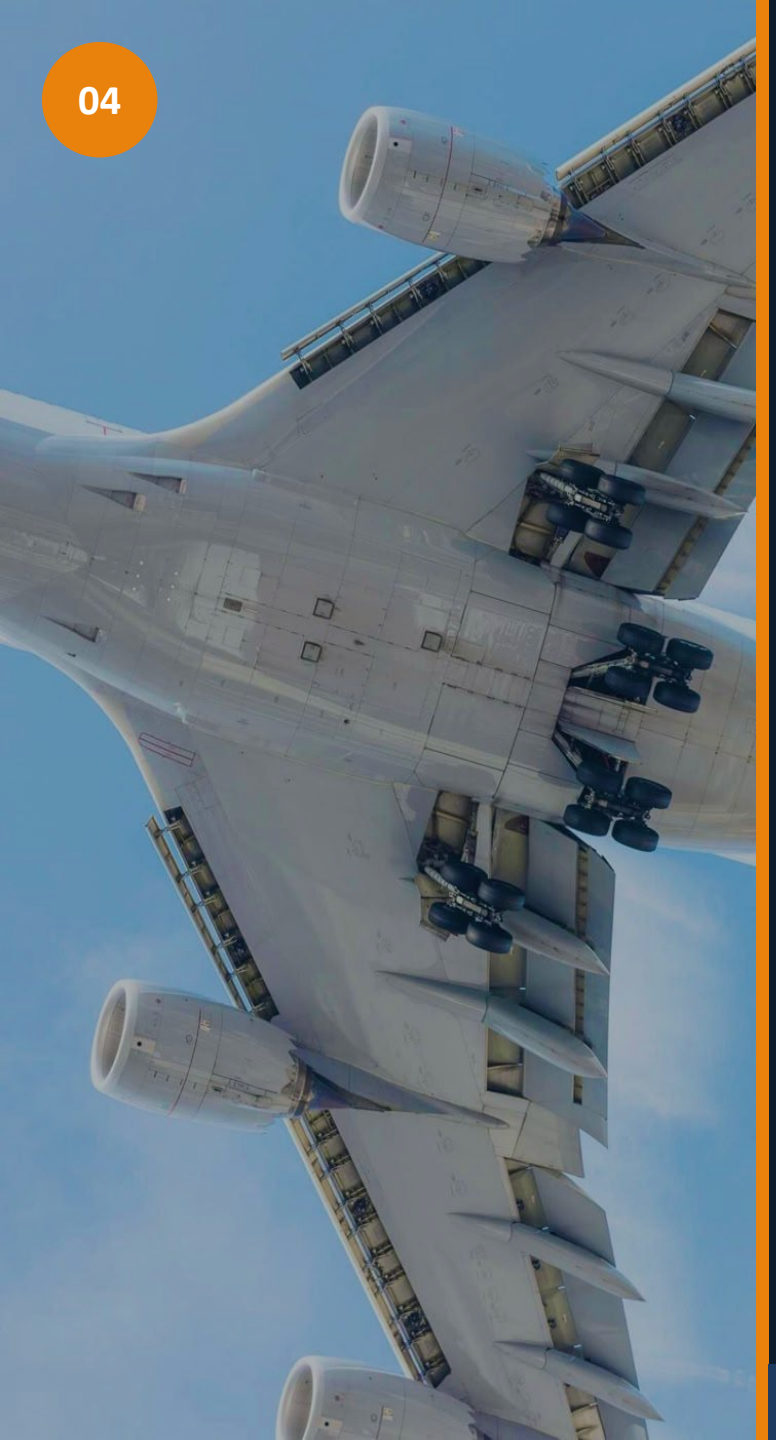
SQL QUERY

```
SELECT a.airline_name,
SUM(ticket_price) revenue
FROM fact_bookings b
JOIN fact_flights f
ON b.flight_id=f.flight_id
JOIN dim_airlines a
ON f.airline_id=a.airline_id
GROUP BY a.airline_name
ORDER BY revenue DESC
LIMIT 1;
```

RESULT

Result Grid | Filter Rows:

airline_name	revenue
Vistara	3805500



Most Popular Routes

Mumbai → Hyderabad

Leads Passenger Demand

Joined flight and airport dimension tables, then grouped by origin-destination pairs to rank routes by booking count.

SQL QUERY

```sql
SELECT
o.city origin,
d.city destination,
COUNT(*) bookings
FROM fact_bookings b
JOIN fact_flights f ON b.flight_id=f.flight_id
JOIN dim_airports o ON f.origin_airport_id=o.airport_id
JOIN dim_airports d ON f.dest_airport_id=d.airport_id
GROUP BY origin,destination
ORDER BY bookings DESC
LIMIT 5;
```

RESULT

Result Grid | Filter Rows:

origin	destination	bookings
Mumbai	Hyderabad	190
Chennai	Bengaluru	174
Kolkata	Chennai	169
Mumbai	Chennai	159
Delhi	Hyderabad	159

Seat Class Demand

Economy: 2,136

Dominates All Classes

Segmented bookings by seat class — Economy leads with 2,136, Business at 1,055, and Premium Economy at 978.

SQL QUERY

```sql
SELECT seat_class,
COUNT(*) total
FROM fact_bookings
GROUP BY seat_class
ORDER BY total DESC;
```

RESULT

Result Grid | Filter Rows:

seat_class	total
economy	2136
Business	1055
ECONOMY	978

Top Customers by Spend

High-Value

Customers Identified

Ranked customers by total ticket spend using SUM aggregation and ORDER BY to surface the top spenders.

SQL QUERY

```
SELECT customer_id,
SUM(ticket_price) spend
FROM fact_bookings
GROUP BY customer_id
ORDER BY spend DESC
LIMIT 5;
```

RESULT

seat_class	total
economy	2136
Business	1055
ECONOMY	978

Airline Delay Performance

SpiceJet

Highest Avg. Delays

Computed average carrier and weather delay per airline using IFNULL to handle nulls, then sorted descending.

SQL QUERY

```
SELECT a.airline_name,
AVG(
IFNULL(weather_delay_min,0)+
IFNULL(carrier_delay_min,0)
) avg_delay
FROM fact_operations o
JOIN fact_flights f
ON o.flight_id=f.flight_id
JOIN dim_airlines a
ON a.airline_id=f.airline_id
GROUP BY a.airline_name
ORDER BY avg_delay DESC;
```

RESULT

airline_name	avg_delay
SpiceJet	42.42028985507246
Air India	40.116279069767444
Vistara	39.08866995073892
IndiGo	37.42339832869081
Akasa Air	37.26878612716763

Customer Segment Revenue

Students: ₹4,447,800

Highest Revenue Segment

Joined customer dimension and fact bookings, grouped by segment — Students generated the most revenue overall.

SQL QUERY

```sql
SELECT segment,
SUM(ticket_price) revenue
FROM fact_bookings b
JOIN dim_customers c
ON b.customer_id=c.customer_id
GROUP BY segment;
```

RESULT

Result Grid | Filter Rows:

segment	revenue
Student	4447800
Business	4104500
Leisure	3805200

Key Business Insights

TOP AIRLINE
Vistara generated the highest airline revenue at ₹3,805,500

SEAT DEMAND
Economy class dominated booking demand with 2,136 bookings

TOP ROUTE
Mumbai–Hyderabad was the most popular route by bookings

CUSTOMER SEGMENT
Student segment produced highest revenue at ₹4,447,800

DELAY PERFORMANCE
SpiceJet showed the highest average delays among all airlines in the dataset